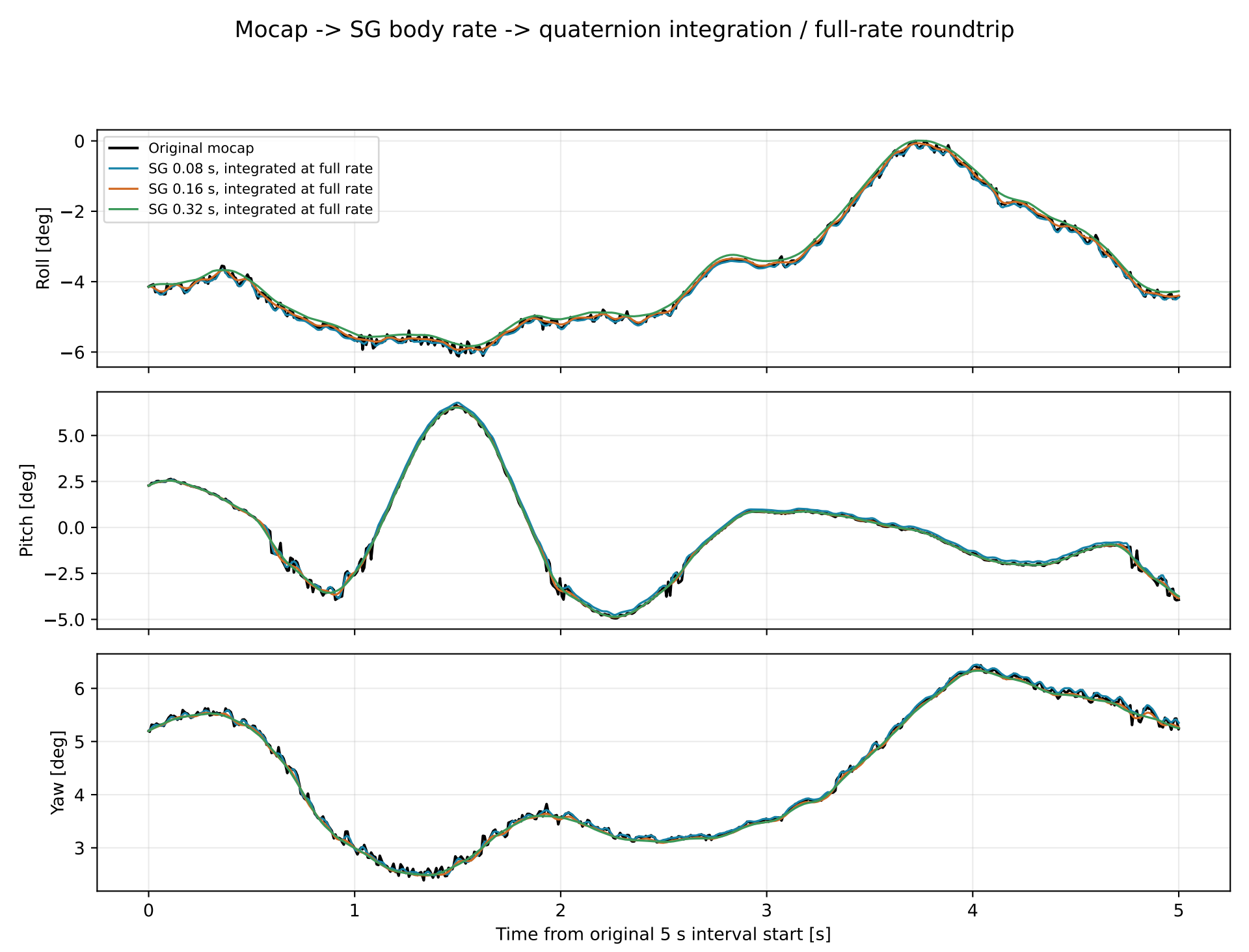
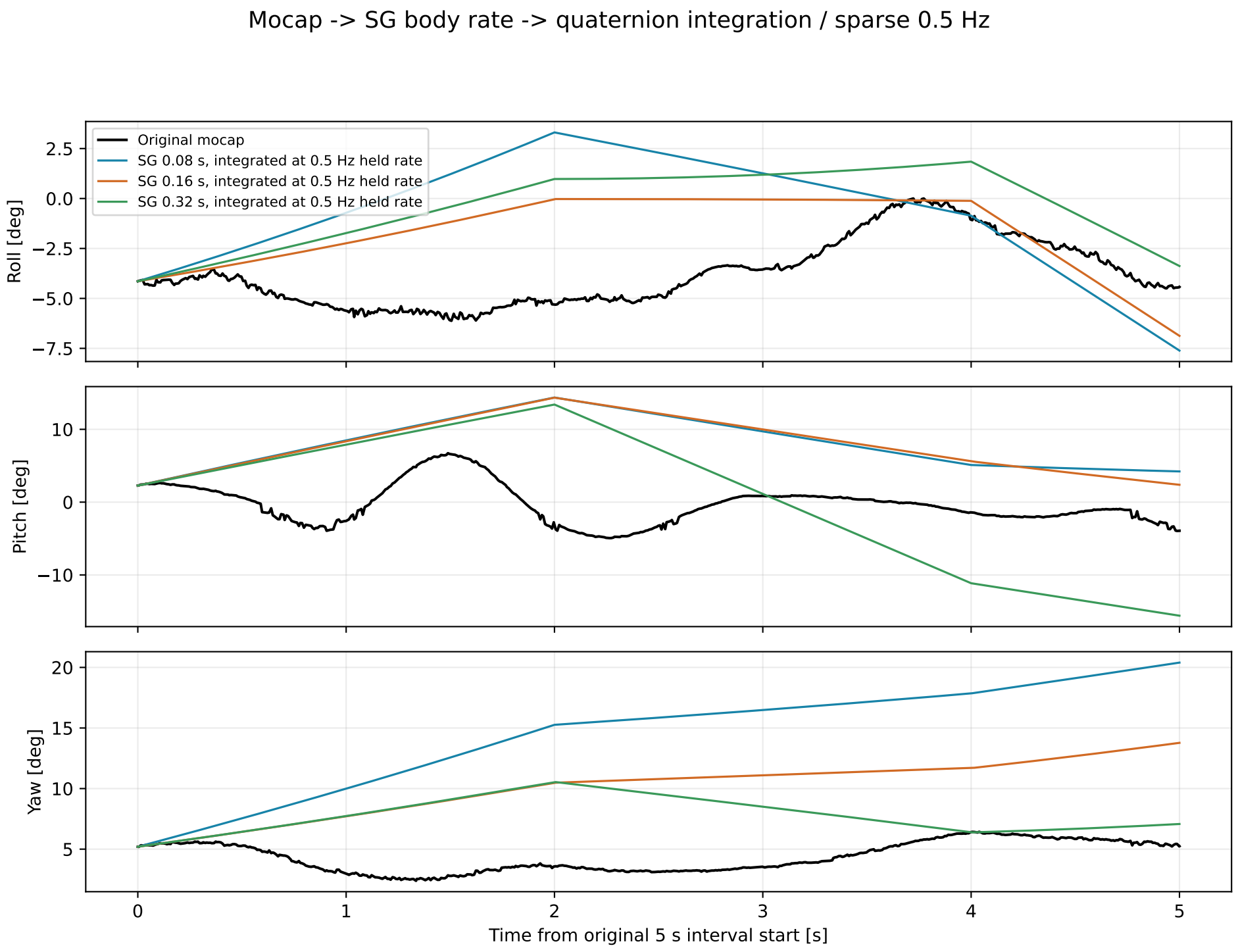
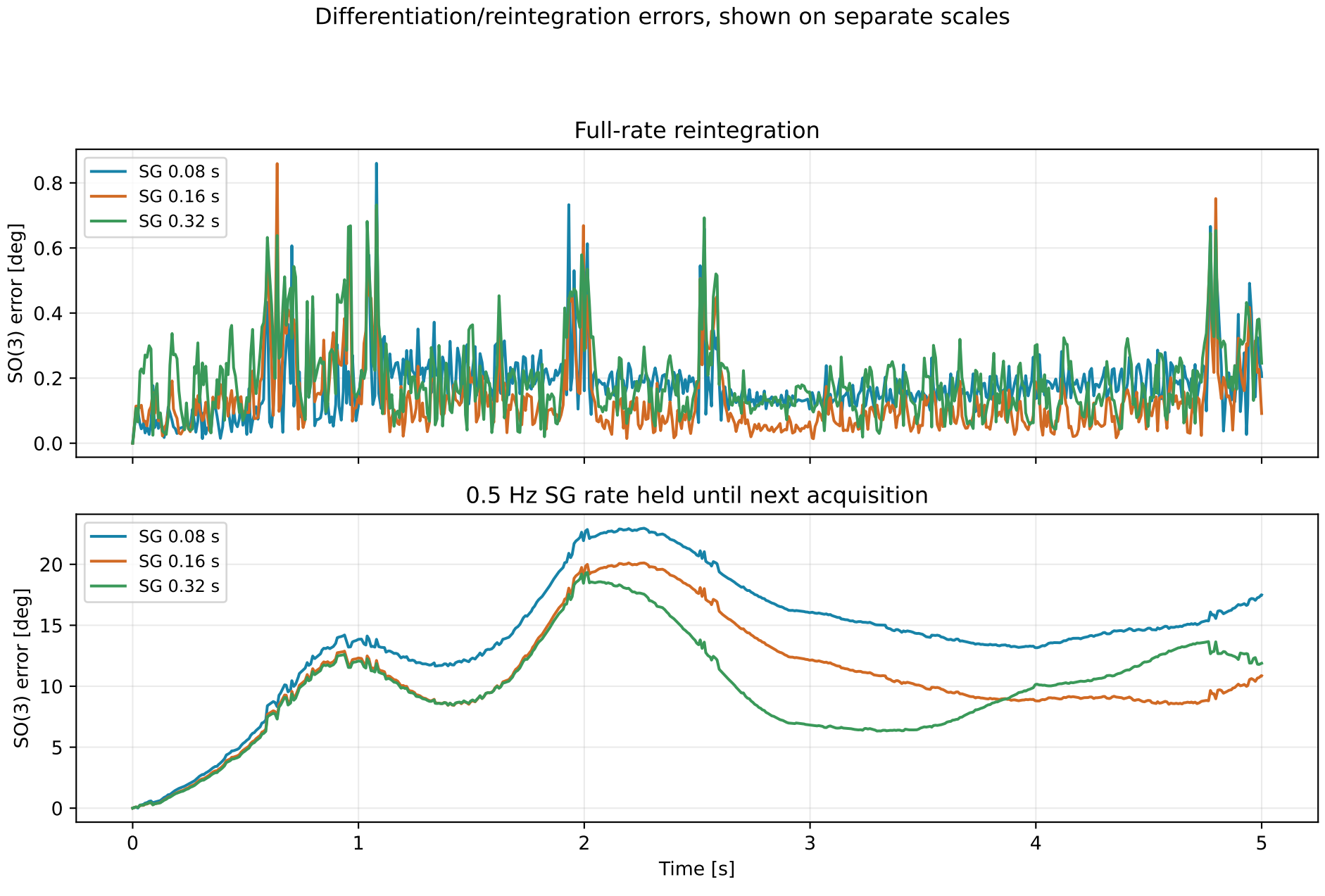

Mocap -> SG body rate -> quaternion integration / full-rate roundtrip
Original mocap
SG 0.08 s, integrated at full rate
SG 0.16 s, integrated at full rate
SG 0.32 s, integrated at full rate
Roll [deg]
Pitch [deg]
Yaw [deg]
Time from original 5 s interval start [s]
Offline consistency diagnostic. Initial mocap orientation used once; no later orientation resets.


Mocap -> SG body rate -> quaternion integration / sparse 0.5 Hz
Original mocap
SG 0.08 s, integrated at 0.5 Hz held rate
SG 0.16 s, integrated at 0.5 Hz held rate
SG 0.32 s, integrated at 0.5 Hz held rate
Roll [deg]
Pitch [deg]
Yaw [deg]
Time from original 5 s interval start [s]
Offline consistency diagnostic. Initial mocap orientation used once; no later orientation resets.


Differentiation/reintegration errors, shown on separate scales
Full-rate reintegration
SG 0.08 s
SG 0.16 s
SG 0.32 s
SO(3) error [deg]
0.0
0.2
0.4
0.6
0.8
0.5 Hz SG rate held until next acquisition
SG 0.08 s
SG 0.16 s
SG 0.32 s
SO(3) error [deg]
0
5
10
15
20
0
1
2
3
4
5
Time [s]